

CHARLES FINNEY YENUMALA

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SUMMARY

Motivated and detail-oriented Computer Science Engineering student with a strong foundation in data structures, algorithms, and core computer science and Artificial Intelligence concepts. Passionate about problem-solving and latest technologies. Skilled in object-oriented programming, full-stack web development, database management. Actively enhancing technical skills through competitive programming, open-source contributions, and internship experience. Seeking challenging roles in software development or related domains to apply and expand my technical expertise in a collaborative environment.

PUBLICATIONS

Predicting SARS-CoV-2 Mutations and Their Impact on Vaccine Efficacy Using Machine Learning Models Trained on RNA Sequencing Data

- Machine learning models trained on RNA sequencing data to predict genetic mutations, enhancing our understanding of their impact on biological functions and disease progression.
- Assess the impact on vaccine efficacy by analyzing genetic sequences, the models aim to identify potential mutations that could affect the virus's behaviour and the effectiveness of existing vaccines.

Intelligent Automation in Software Engineering: A Multi-Modal AI Framework for Proactive Bug Detection and Code Correction

- Built a multi-modal AI framework for proactive bug detection and automated code correction, integrating static analysis (SonarQube, Bandit), dynamic testing, machine learning (Random Forests, SVM), and NLP (Transformer models) to enhance code reliability.
- Leveraged AST parsing, cyclomatic complexity metrics, and semantic analysis to achieve high Precision, Recall, and F1-Score in realworld evaluations, improving developer productivity and reducing defect rates.

TECHNICAL SKILLS

Core Knowledge Areas: Artificial Intelligence, Machine Learning, Deep Learning, AWS, Figma, Data Structures, Object Oriented Programming, Database Management

Programming Languages: C, Python, Java, SQL, HTML, CSS

Databases: MongoDB, Postgres

PROFESSIONAL EXPERIENCE

INTERN | Pramahasoft Solutions Pvt Ltd - Software Developer Intern (Dec 2025 – May 2026)

Project: UTELO (Real-Time Voice & Chat Translation Platform)

Full-Stack Mobile Development:

- Developed a cross-platform mobile application using Flutter for the frontend and Node.js/WebSocket for the backend, enabling seamless real-time communication between users speaking different languages.
- Designed and implemented a custom WebSocket-based signaling server to handle instant message delivery, call connection states, and live audio streaming with sub-second latency.

Cloud Architecture & API Migration:

- Led the critical migration of the entire translation pipeline from Azure Cognitive Services to Google Cloud Platform (GCP) to optimize costs and improve translation accuracy for Indian regional languages.
- Integrated Google Speech-to-Text (STT), Google Translate, and Google Text-to-Speech (TTS) APIs into a unified backend pipeline, reducing dependency on third-party SDKs.

Real-Time Audio Engineering:

- Solved complex "echo" and "feedback loop" issues in full-duplex voice calls by implementing event-driven audio handling and microphone/speaker state management in Flutter.
- Implemented robust error handling for real-time streams, wrapping critical "Room Logic" in safety blocks to prevent backend crashes during user disconnects.

Deployment & DevOps:

- Deployed and managed the Node.js backend on Render.com, configuring environment variables, build commands, and health checks for continuous availability.

Key Technologies:

- Flutter, Dart, Node.js, Express, WebSockets (ws), Google Cloud APIs (STT/TTS/Translation), Firebase (Auth/Firestore), Render.com.

INTERN | Code Alpha, Code Soft, Bhart Internship

May 2024 – Aug 2024

- Worked on projects based on AI and ML
- Designed and implemented a multilingual chatbot enabling real-time translation across Indian languages for improved user accessibility.
- Developed custom NLP pipelines to handle colloquial expressions, idioms, and context-aware translations, enhancing conversational accuracy.

EDUCATION

BACHELOR'S DEGREE | Computer Science and Engineering | KL University | CGPA: 8.2 2022 – 2026

INTERMEDIATE DEGREE | MPC | Vikas the Concept School | CGPA: 7.0 2020 – 2022

SECONDARY EDUCATION | Vikas the Concept School | CGPA: 7.3 2019 – 2020

CERTIFICATIONS

- MongoDB Node.js Developer Path Feb 23, 2024
- Computer Networking Course: Master Computer Networking Mar 20, 2024
- Postman API Fundamentals Student Expert Mar 22, 2024

ACHIEVEMENTS

- Represented India in Singapore Fins International in swimming.
- Indian National Bronze medalist in 200M freestyle swimming.
- Senior National Aquatic Championships (participant)
- Junior National Aquatic Championships (participant)
- All India University Games (participant)
- South and west zone University Games (Top 20)
- Fines National Aquatic Championships 2022, 2023, 2024, 2025 (22 Gold, 8 Sliver, 5 Bronze medals)
- Fip sliver international padel Tournament (Top 16 in 2025)
- Overall, Champion at BITS Pilani University Sports Meet (BOSM) 2024 and 2025 (10 golds medals)
- Overall, Champion Mahindra Sports & Gaming Events 2024 (6 golds medals)

VOLUNTEERING

- Member of Ivarna Club at KL University
- Conducted several blood donation camps at Vasavi Hospital
- Treasurer of Sports Club at KL University
- Member of Singing club (beat boxing) at KL University

INTERESTS

Swimming | Paddle | Pickleball | Badminton | Traveling | Gym | Coding